

Why are AHP results not binding for appraising real estate?

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ABSTRACT

Using the method of reduction to absurdity, an initial hypothesis was considered: the results of the general weighting vector of AHP are not linked to real estate market prices, as opposed to the alternative hypothesis that they are linked. To demonstrate this, a "real fable" was used to explain the valuation of a medieval castle. The first example was based on three criteria: quality of the environment, quality of the castle, and state of conservation; and six alternatives, five comparables plus the property being appraised (3C+6A), completing three more cases. Using AHP, the elements of the global vector of the alternatives were determined. With the offer prices and the use of Valuation Ratio (RV), the value of the property and the values of the comparables were estimated for comparative analysis. The Non-Ratio Reciprocity Test (NRRT) was used, obtaining significant percentage differences between the values and prices of the comparables, not guaranteeing reciprocity to validate the initial estimates of the appraised property value. In the second example, three different criteria were used: strength, security, and access; the same six alternatives were used (3C+6A). It was possible to demonstrate the inconsistency of the results due to the disparity of the values obtained in all cases of the two examples for the appraised property, which does not allow for guaranteeing reliable market values. For the third example, the database simulation was performed based on the information from the original sample, with 8 explanatory variables, to obtain 69 data points. Econometric modeling was carried out, demonstrating that there is no correlation between the explained variable (unit price) and the explanatory variables as a whole, much less individually, therefore, the results are not reliable. Despite the non-fulfillment of the basic assumptions of MRLM, point and interval estimation was done. These values were similar to those obtained in the second example, using AHP+RV. With this evidence, the analyzed proposal does not guarantee market values; therefore, the initial hypothesis is correct.

Key words: AHP, models, valuation, real estate, MRLM.